

- 5 The Chamorro people of Guam eat flying foxes, a type of bat, as part of their diet.

The photograph shows a flying fox.



(Source: © H Lansdown/Alamy Stock Photo)

Scientists in Guam suggested there was a link between the diet of these people and the high incidence of the neuro-degenerative disease amyotrophic lateral sclerosis (ALS) in their population.

- (a) In one investigation, scientists looked at the accumulation of the neurotoxin BMAA in the food chain of the Chamorro people.

The table shows some of the data.

Tissue	Concentration of BMAA / μg per gram
Seeds of cycad trees	37
Fat of flying fox	3 556

Calculate the percentage increase in BMAA per gram from the seeds of the cycad trees to the flying foxes.

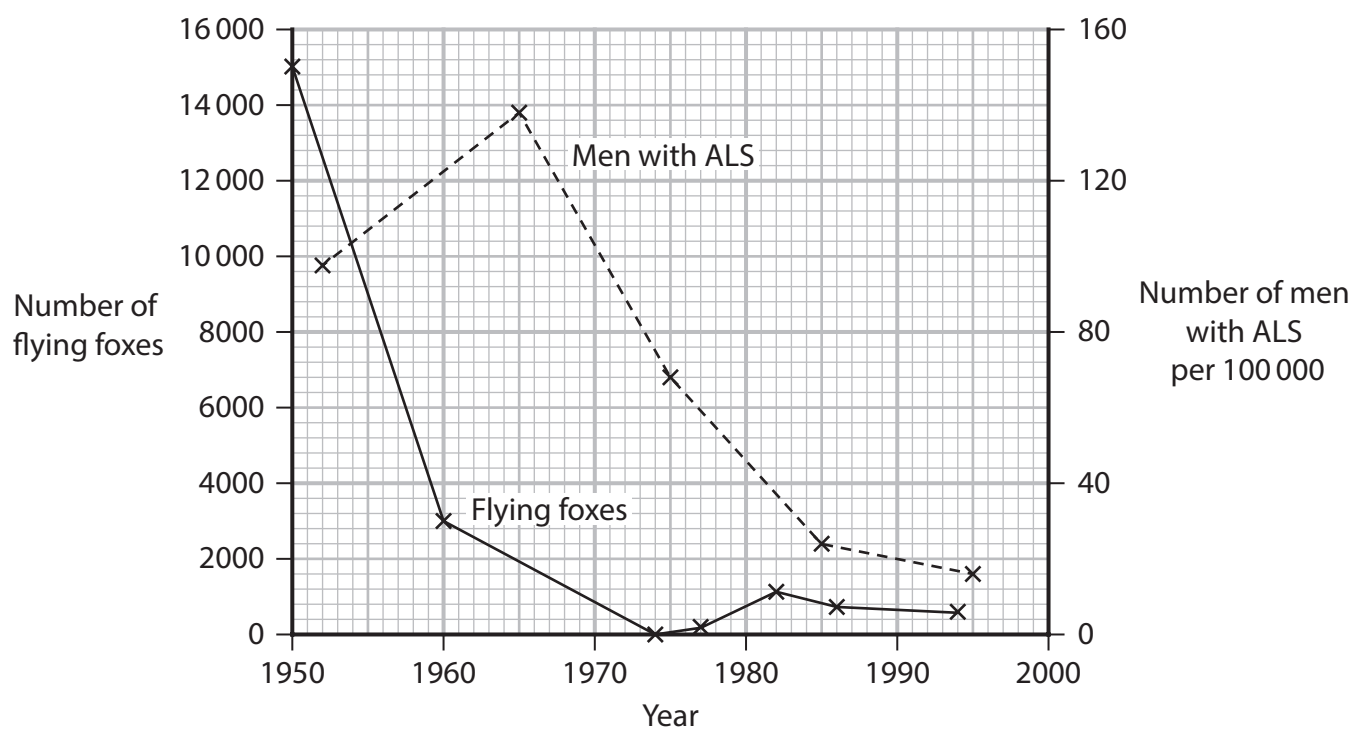
Give your answer to **two** significant figures.

(1)

Answer%

- (b) In a second investigation, the scientists looked at the flying fox population and the number of men diagnosed with ALS.

The graph shows the results of this investigation.



(i) Explain the results of this investigation.

Use information from the question to support your answer.

(4)

(ii) The diagnosis of ALS is made using positron emission tomography (PET) scans and computed tomography (CT) scans of the brain and nervous tissue.

Compare and contrast these two techniques.

(3)

[illegible]

(c) Neurotoxins can affect the potential differences generated across the membranes of motor neurones.

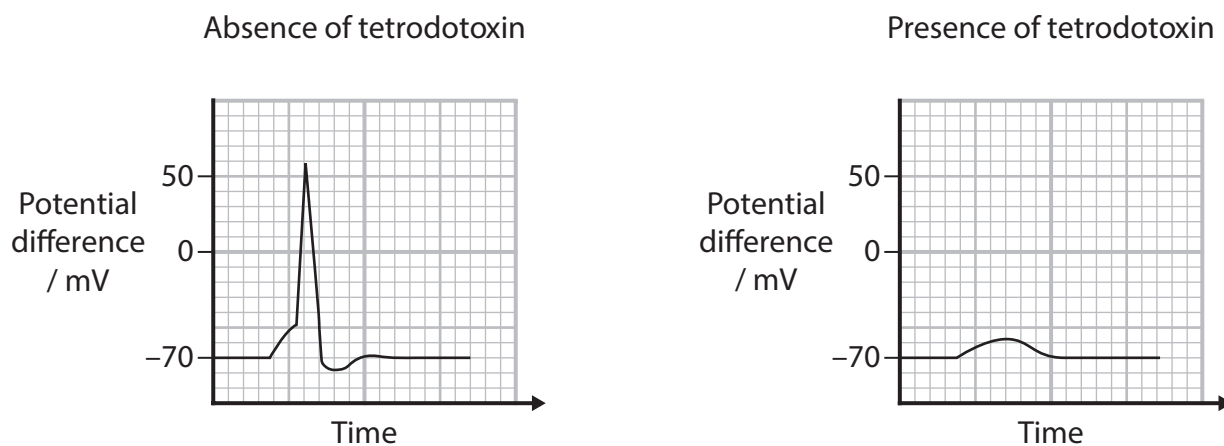
(i) Draw a myelinated motor neurone and label **two** structures.

(3)

- (ii) A different neurotoxin called tetrodotoxin inhibits the transmission of nerve impulses along axons.

In an experiment, axons of motor neurones were stimulated in the absence of tetrodotoxin and in the presence of tetrodotoxin.

The graphs show the potential differences obtained in this experiment.



Explain how tetrodotoxin inhibits the transmission of a nerve impulse along the axon.

(3)

(Total for Question 5 = 14 marks)